

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination, 2023-24

Paper: CS-101 — Problem Solving through C Programming

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1 which is compulsory. Write your Roll No. at the top immediately on receipt of this question paper.*

Question 1. Attempt any *nine*.

[2×9 = 18]

- (a) Differentiate between `float` and `double` data types in C.
- (b) What is the difference between the `=` operator and the `==` operator in C?
- (c) What is the role of the `break` keyword in C programming?
- (d) What is the difference between a string and an array of characters?
- (e) What is the use of the `#ifndef` preprocessor directive in C programming?
- (f) What is the difference between `scanf()` and `gets()` functions?
- (g) Write the syntax and usage of the `->` operator in C.
- (h) Define the concept of *pointer to pointer* with a suitable example.
- (i) What will be the output of the following code? (Explain with reason.)

```
void main() {  
    int x = 10, y = 5, p, q;  
    p = x - 19;  
    q = (p++) && (y = 3);  
    printf("p=%d y=%d", p, q);  
}
```

- (j) What will be the output of the following code? (Explain with reason.)

```
void main() {  
    int x = 3, y = 2, z;  
    z = x++ + ++y;  
    printf("x=%d y=%d z=%d", x, y, z);  
}
```

Question 2.

- (a) What is an *algorithm*? Draw the flowchart to print the real roots of the quadratic equation $ax^2 + bx + c = 0$ using the Sridharacharya formula. [4]
- (b) Explain the difference between a `while` loop and a `do...while` loop with a suitable example. [5]

- (c) Explain the `switch...case` statement with a suitable example. [4]

Question 3.

- (a) Discuss the *logical operators* in C with a suitable example. [4]
- (b) Differentiate between *actual* and *formal* arguments when using a function. Explain *call by reference* with a suitable example. [5]
- (c) Define *recursion*. Write a program to find the factorial of a given number using recursion. [4]

Question 4.

- (a) State the difference between `register` and `static` storage class variables with appropriate examples. [4]
- (b) Write your own functions in C for the following string operations: [6]
- (i) Copy one string into another.
 - (ii) Reverse a given string.
- (c) What are *command line arguments*? Why are they used? [3]

Question 5.

- (a) How are elements of a 2D array stored in computer memory? Give one example. [2]
- (b) Differentiate between the concepts of *pointer to an array* and *array of pointers* with suitable examples. [6]
- (c) How can a pointer be used to access the elements of a one-dimensional array? Explain with an example. [5]

Question 6.

- (a) Explain the different possible *modes* for opening a file in C. Write a program to copy the contents of one file into another. [5]
- (b) What do you mean by *dynamic memory allocation* in C? Explain the role of the `realloc()` function with a suitable example. [4]
- (c) Define *structure* with a suitable example. How is memory allocation done for structure variables? Give one example. [4]

Question 7.

- (a) What is the role of the *Preprocessor* in C programming? Differentiate between the two forms of include directives: `#include<mylib.h>` and `#include"mylib.h"`. [4]
- (b) Explain the concept of *nested structure* with a suitable example. [5]

- (c) Define *union* with a suitable example. Give one real-life example where a union can be used. [4]